

ELKHORN SLOUGH RESERVE, CA, USA

WMO: 997997

Lat: 36.815N

Lon: 121.738W

Elev: 3

StdP: 101.29

Time Zone: -8.00 (NAP)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	1.8	3.0	-5.0	2.5	9.7	-2.8	3.0	8.8	8.4	12.9	7.4	12.9	0.9	50	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	7.3	25.3	16.7	23.3	16.0	21.5	16.3	19.1	21.4	18.4	20.8	17.7	20.1	2.5	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	13.3	19.8	17.6	12.6	19.2	16.7	11.9	18.4	54.2	20.9	52.0	20.8	49.6	20.1	22.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.4	5.9	5.2	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.3	11.8	10.9	11.6	11.9	12.9	14.3	15.2	15.7	15.9	15.5	12.9	10.6
	DBStd	3.03	3.19	2.98	2.37	2.16	1.82	1.66	1.56	1.83	2.12	2.63	2.85	2.66
	HDD10.0	87	19	22	9	7	0	0	0	0	0	0	6	24
	HDD18.3	1867	204	208	209	195	168	123	97	84	79	95	164	240
	CDD10.0	1284	74	47	59	62	91	128	162	177	176	171	93	43
	CDD18.3	23	0	1	1	1	1	2	1	2	5	8	2	0
	CDH23.3	168	3	4	7	5	11	10	4	7	34	67	17	0
	CDH26.7	30	0	0	2	0	1	4	1	0	6	16	1	0
Wind	WSAvg	2.1	2.1	2.1	2.2	2.6	2.7	2.5	2.4	2.1	1.9	1.8	1.7	1.8
Precipitation	PrecAvg	444	96	92	63	32	14	2	0	1	1	22	40	86
	PrecMax	733	245	325	237	94	51	14	1	7	10	102	132	199
	PrecMin	233	0	0	0	3	0	0	0	0	0	0	0	2
	PrecStd	122	71	70	62	27	16	5	0	2	3	27	36	54
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.7	23.7	23.4	23.7	24.9	26.0	22.3	24.7	27.0	28.3	25.9	20.0
		MCWB	13.3	14.6	15.6	15.8	15.8	16.2	18.5	18.3	17.9	16.0	15.3	14.1
	2%	DB	21.1	20.1	19.2	19.6	19.7	20.6	20.7	21.9	23.6	25.0	22.8	18.0
		MCWB	13.8	13.3	14.3	14.8	14.2	16.4	17.8	18.3	17.2	16.1	14.4	13.5
	5%	DB	19.0	17.3	17.5	17.3	17.7	18.8	19.5	20.6	21.2	22.1	19.5	16.3
		MCWB	12.4	13.0	13.8	13.6	14.3	15.8	17.4	17.6	17.2	16.3	13.2	13.0
	10%	DB	16.7	15.5	15.8	15.8	16.4	17.6	18.3	19.2	19.8	20.1	17.5	15.0
		MCWB	11.8	12.2	12.9	12.7	13.7	15.2	16.3	16.8	16.6	15.8	12.8	12.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.9	16.3	17.0	17.1	16.7	18.8	19.9	20.2	19.1	20.0	16.7	16.8
		MCDB	19.7	19.9	21.4	22.4	21.5	20.4	21.7	23.2	22.9	22.6	21.9	18.4
	2%	WB	14.5	15.0	15.5	15.2	15.5	17.4	18.7	19.0	18.4	18.6	15.5	15.6
		MCDB	17.4	17.9	18.2	18.3	18.6	19.7	19.8	21.0	22.0	21.2	19.2	16.7
	5%	WB	13.6	13.8	14.4	14.0	14.7	16.2	17.7	18.0	17.7	17.6	14.7	14.6
		MCDB	16.9	16.6	16.6	16.5	17.2	18.4	19.1	20.0	20.7	20.2	17.6	15.3
	10%	WB	12.8	12.7	13.3	13.1	14.0	15.2	16.6	17.0	16.9	16.6	13.8	13.4
		MCDB	16.0	15.0	15.4	15.0	16.2	17.2	18.0	18.9	19.5	19.1	16.4	14.2
Mean Daily Temperature Range	5% DB	MDBR	9.8	9.0	8.9	8.3	7.2	6.5	5.5	5.9	7.3	9.5	9.9	8.8
		MCDBR	12.7	11.8	11.7	11.5	10.1	9.0	7.3	8.3	10.5	13.3	13.1	9.9
		MCWBR	7.0	6.7	7.1	7.2	6.2	5.4	4.5	4.9	6.0	7.0	6.6	6.0
	5% WB	MCDBR	10.0	9.7	10.5	9.8	8.5	8.0	6.8	8.0	9.3	9.7	10.9	6.8
		MCWBR	6.2	6.1	6.8	6.5	5.5	5.5	4.6	4.9	5.5	6.3	6.2	5.1
Clear-Sky Solar Irradiance	taub	0.320	0.325	0.326	0.352	0.363	0.354	0.357	0.341	0.350	0.323	0.311	0.315	
	taud	2.546	2.539	2.536	2.438	2.422	2.458	2.470	2.503	2.471	2.579	2.580	2.575	
	Ebn at Noon	875	915	944	929	918	923	918	928	904	905	881	859	
	Edh at Noon	78	88	96	112	116	112	110	104	101	83	74	71	
All-Sky Solar Radiation	RadAvg	2.58	3.47	4.89	6.22	7.20	7.86	7.41	6.68	5.70	4.25	3.00	2.28	
	RadStd	0.29	0.38	0.41	0.52	0.41	0.31	0.21	0.24	0.20	0.25	0.16	0.29	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (40 neighbors)	+0.40	N/A	-1.20	N/A	N/A	N/A	N/A	N/A	-120	+146	+36

Nomenclature: See separate page